

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

MARK SCHEME

CANDIDATE
NAME

CLASS

BIOLOGY

Paper 3

9747/03

September 2009

1 hour 30 minutes

Additional Materials: Answer Booklet / Paper

READ THESE INSTRUCTIONS FIRST

Write your name, and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
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4	
Total	

This document consists of **12** printed pages and **no** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 Scientists have recently produced genetically-engineered clover containing a protein high in sulphur-containing amino acids. As proteins in wool are also rich in these amino acids, it is thought that feeding sheep on this new strain of clover (leaves are consumed) will produce an increase in the yield of wool. A DNA fragment was prepared from the sequences shown in **Fig. 1.1** for insertion into the clover.

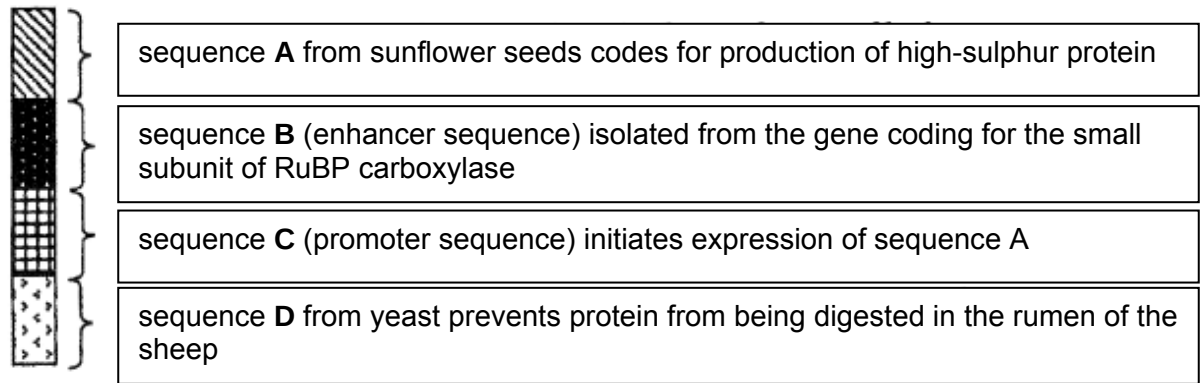


Fig 1.1

- (a) Describe how enzymes are used to construct the DNA fragment shown in **Fig. 1.1**.

Restriction enzymes are used to synthesize complementary sticky ends between each fragment;

DNA ligase is used to form phosphodiester bonds between nucleotides / fragment;

[2]

- (b) Suggest why sequence **B** was included in the DNA introduced into the clover.

Increase the expression of sequence A;

In photosynthesizing cells which are in the leaves of the plant;

[2]

- (c) Fig. 1.2 shows a vector map of the plasmid vector pSING003. The plasmid is used to clone the DNA fragment, which will be inserted at the *EcoRI* site. The plasmid is subsequently introduced into *E. coli* via transformation. After transformation, the *E. coli* cells are then grown on an agar plate.

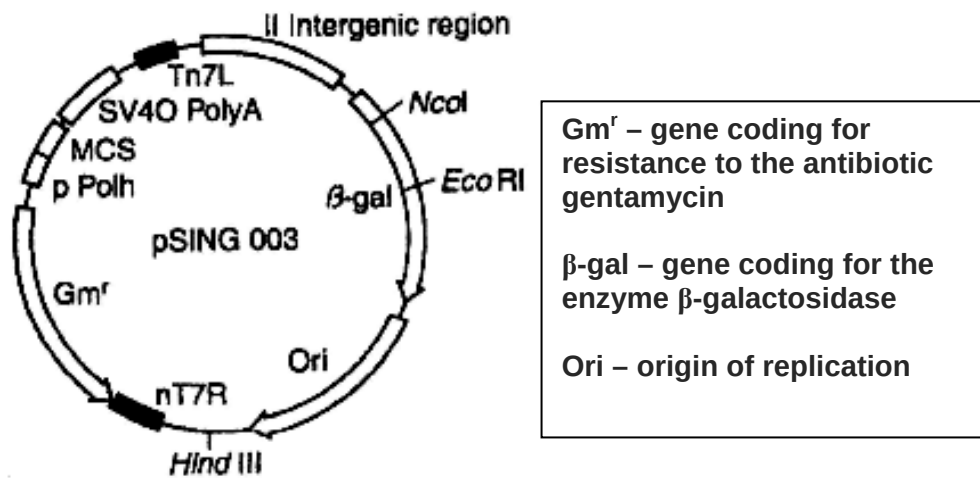


Fig. 1.2

- (i) Describe **two** properties of plasmids that allow them to be used as DNA cloning vectors.

Possess own origin of replication thus can replicate independently of main chromosome;

Possess restriction sites which can be cut by restriction enzymes for insertion of gene-of-interest;

[2]

- (ii) In addition to the nutrients required for growth, name **two** other substances that are added to the agar plate to allow selection of cells successfully transformed with the recombinant plasmid.

1. X-gal;

2. Antibiotic gentamycin;

[2]

- (d) Fig. 1.3 shows the results of plating the transformed cells onto an agar plate with the appropriate substances.

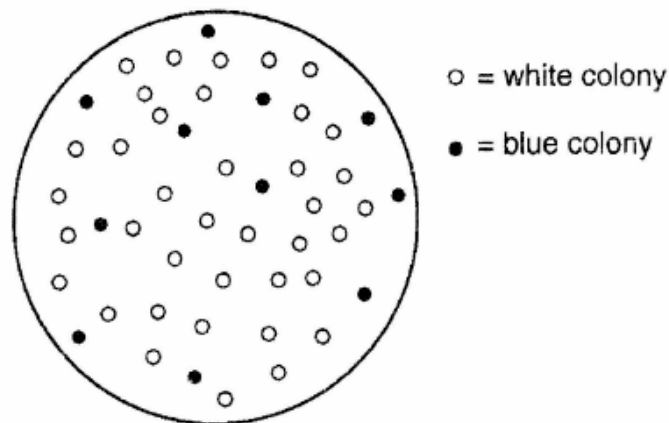


Fig. 1.3

- (i) Explain why some colonies appeared white while others appeared blue.

Intact β -galactosidase gene codes for enzyme which hydrolyzes X-gal, giving rise to blue colour;

Bacteria colonies with reannealed plasmids will appear blue;

Bacteria colonies with recombinant plasmids will appear white;

Due to insertional inactivation which disrupts β -galactosidase gene;

[4]

- (ii) Using Fig. 1.3, calculate the percentage of cells into which the recombinant plasmid had been successfully introduced. Show your working in the space provided below.

Percentage of cells = $38/49 \times 100 = 77.6\%$ (correct to 3 sig. fig.);

[1]

- (iii) Suggest why replica plating was not necessary in this experiment.

Replica plating is needed only if the bacteria colonies which possess recombinant or reannealed plasmids are cultured on the same medium and cannot be differentiated;

In this case, colonies with recombinant plasmids can be differentiated visually by their colours;

[2]

[Total: 15]

- 2 **Fig. 2.1** summarises the large-scale production of three proteins of medical importance, insulin, human growth hormone (HGH) and antithrombin. **P** and **Q** are enzymes.

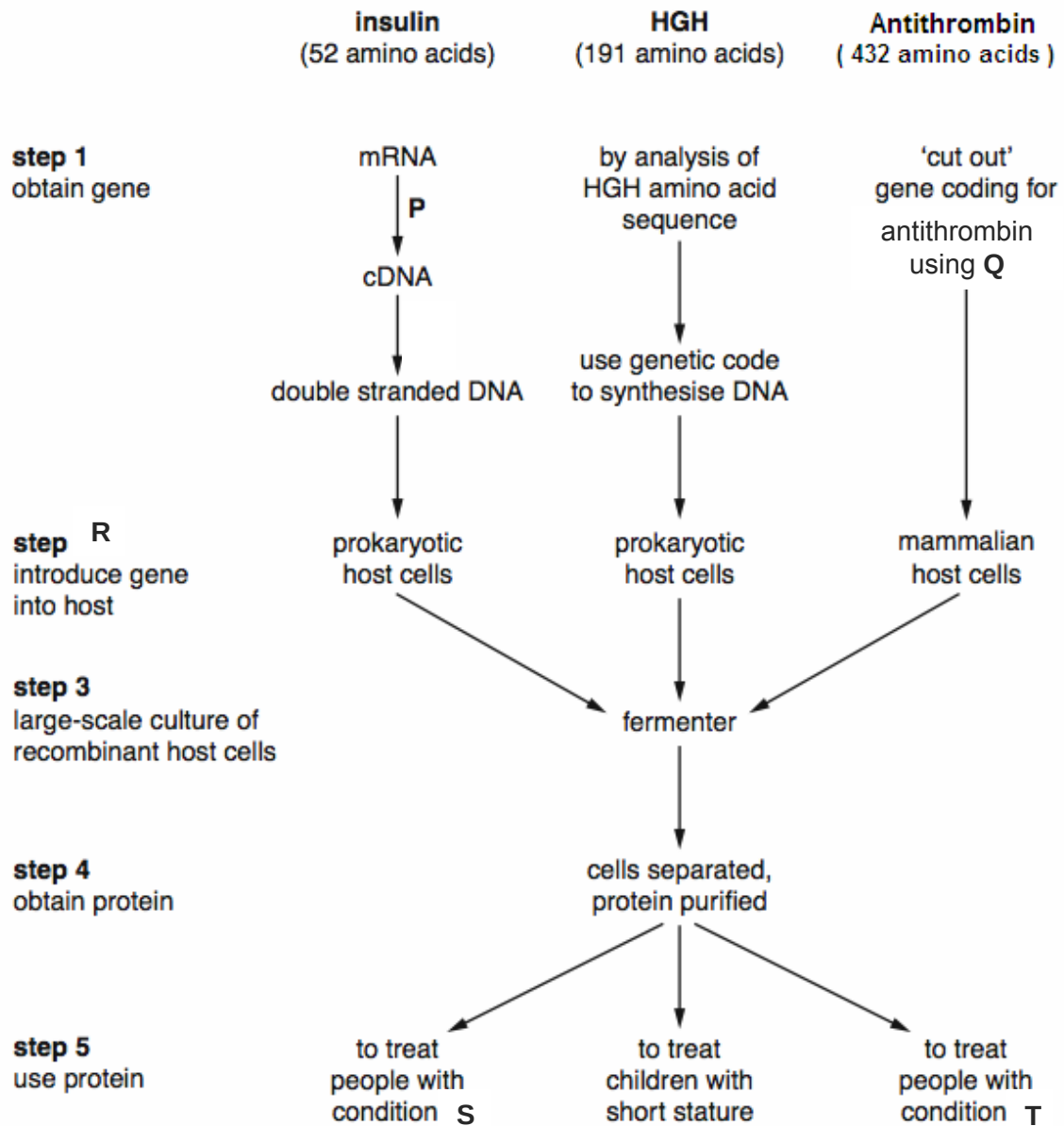


Fig. 2.1

- (a) From **Fig. 2.1**, name the enzymes **P** and **Q**, step **R** and conditions **S** and **T**.

Enzyme P	reverse transcriptase	}	Condition S	diabetes	}
Enzyme Q	restriction enzyme		Condition T	thrombosis	
Step R	transformation;			@ anti-thrombin deficiency	

[3]

- (b) Suggest why the method used to synthesise the gene coding for HGH is not used to obtain the gene coding for antithrombin.

HGH much larger than antithrombin, so may not be possible to synthesize the DNA fragment artificially;

[1]

- (c) Antithrombin deficiency (AD) is a blood clotting disorder in which sufferers produce insufficient amounts of antithrombin. Antithrombin binds to thrombin and heparin resulting in deactivation of the blood coagulation pathway. AD results in formation of blood clots in blood vessels which obstruct blood flow in the circulatory system.

RFLP analysis can be used as a method of detecting the disease in family members by using a radioactive probe to determine the presence of the mutant allele that results in AD. Fig. 2.2 shows a pedigree diagram and an autoradiogram of a family with history of AD.

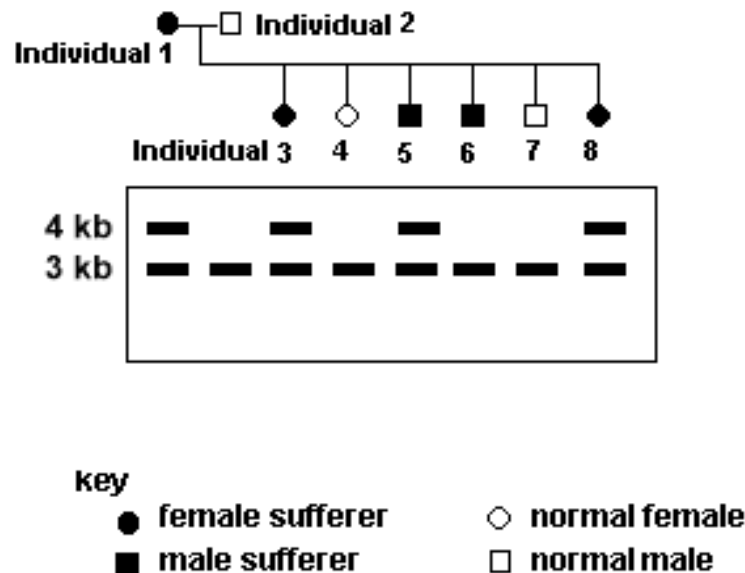


Fig. 2.2

- (i) With reference to Fig 2.2, explain which RFLP fragment is linked to the disease-causing AD allele.

4kb fragment;

All diseased individuals (except 6) / Individuals 1, 3, 5 and 8, possesses the 4kb fragment;

indicates that the RFLP probe binds to a region very close to the mutant / disease allele;

[3]

- (ii) Suggest why some individuals possess only one band.

Homozygous condition / possess two identical copies of the allele;

[1]

- (iii) Explain the mode of inheritance of AD.

Dominant because equal chances of males and females inheriting the disease;

Autosomal because only one copy of 4kb fragment required to cause disease;

[2]

- (iv) On Fig. 2.3, draw the location of the probe that would produce an autoradiogram as indicated in Fig. 2.2. [1]

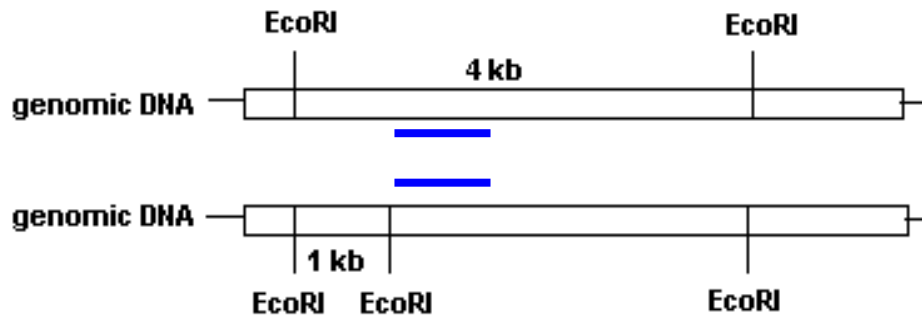


Fig. 2.3

- (v) On Fig. 2.4, draw the electrophoretic gel pattern of individuals 1 and 2. [1]

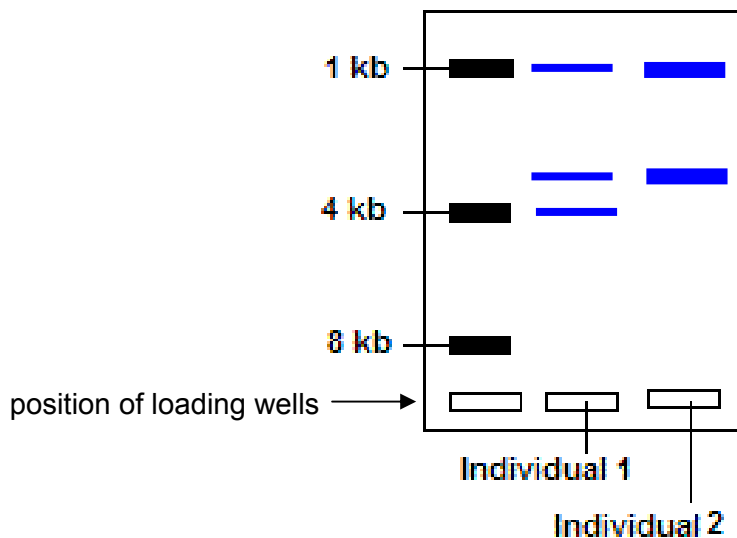


Fig. 2.4

- (d) Explain the limitations of gene therapy that prevent it from being used as a treatment for patients suffering from AD.

Short-lived nature of gene therapy so patients have to continually undergo therapy;

Incorrect insertion of therapeutic gene resulting in other diseases;

Triggering of immune response resulting in rejection;

Problems with finding suitable viral vectors due to unknown toxicity of these viruses;

[3]

[Total: 15]

- 3 An investigation was performed into the possible spread of a transgene into maize grown in the remote mountains of Southern Mexico. Southern Mexico is the region of origin and the centre of diversity of maize. Mexico has banned the planting of transgenic maize since 1998.

Grains of native landraces of maize grown in two sites were tested for the presence of a DNA promoter sequence derived from cauliflower mosaic virus (CMV). This promoter is commonly used in transgenic maize.

Maize grains from a 30-year old museum collection in Mexico and blue maize from remote mountains in Peru were also tested for the presence of CMV promoter sequence.

The results of the investigation are shown in **Table 3.1**.

Type of maize	Existence of CMV promoter DNA sequence
Mexican native landrace maize – site 1	Present in 3 – 10% of grains
Mexican native landrace maize – site 2	Present in 3 – 10% of grains
Transgenic maize	Present in 100% of grains
Maize from museum collection in Mexico	Absent
Blue maize from Peru	Absent

Table 3.1

- (a) Using the information given and the results in **Table 3.1**, explain

- (i) why grains from the museum collection in Mexico and blue maize from Peru were both tested for the presence of CMV promoter DNA.

Used as a control / point of reference that CMV promoter DNA was not naturally occurring in native maize plants;

Testing these grains formed a control as the museum collection was collected before any genetic engineering;

While blue maize from Peru was from a remote region, thus unlikely to have been contaminated by pollens from transgenic plants;

[3]

- (ii) how CMV promoter DNA has been passed to native landrace maize.

Pollen grains containing CMV promoter from transgenic plants may be transferred by wind pollination to native plants;

Cross pollination by pollen grains containing the transgene and its CMV promoter, with native plants may give rise to generations of plants containing CMV promoter;

[2]

- (b) Suggest why the CMV promoter DNA is present in native landrace maize two years after the ban on cultivation of transgenic maize in Mexico.

Smuggling of transgenic maize by plants breeders;

Presence of residual transgene in subsequent generations;

[1]

In a separate experiment on the impact of transgenic crop plants on the natural environment, the survival of transgenic and non-transgenic crop plants was calculated. The mean percentage survival of the plants in different habitats at the end of the first growing season is shown in **Fig. 3.2**.

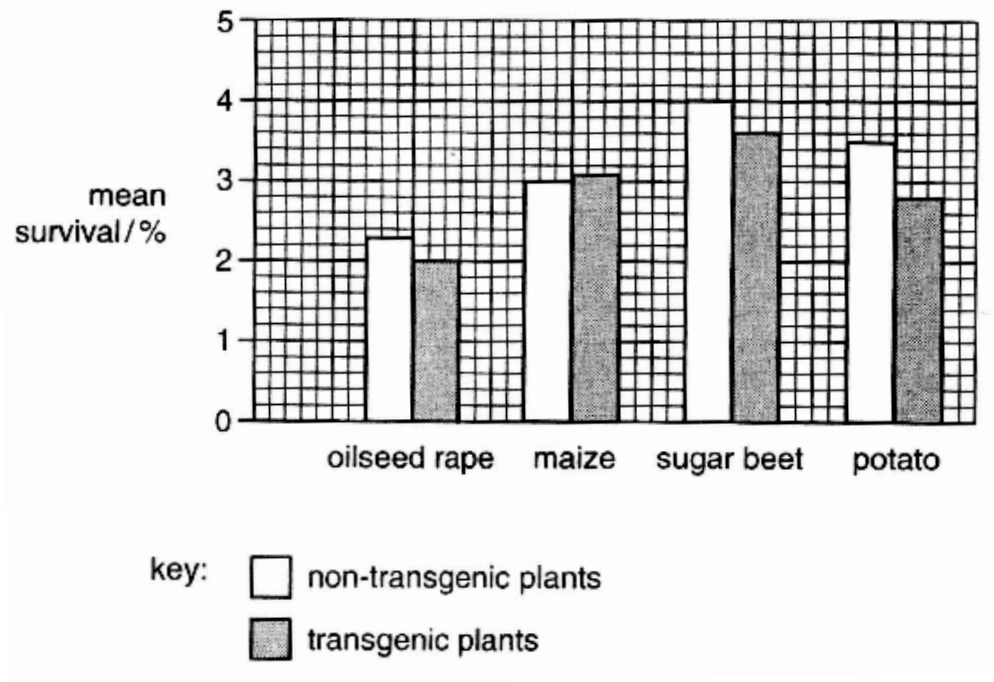


Fig. 3.2

- (c) Using the data in **Fig. 3.2**, compare the mean percentage survival in natural habitats of transgenic and non-transgenic crop plants at the end of the first growing season.

Not significantly different in survival rates between transgenic and non-transgenic crops;

Both have relatively low survival rate of between 2 – 5%;

Only in maize is survival of transgenic crop greater than non-transgenic by 0.1%;

Potato has greatest difference in survival between transgenic and non-transgenic by 0.7%;

[4]

Besides genetically modifying crop plants, animals have also been genetically engineered for various purposes. Salmon growth hormone (sGH) has been inserted into plasmids and the recombinant plasmids were microinjected into eggs from very slow-growing domestic and wild strains of Atlantic salmon.

Their growths were scored, as shown in **Fig. 3.3**. In a further analysis, growth hormone was extracted from domestic-GH treated salmon on dates 11/5, 23/6, 14/7 and 1/9. A protein gel was run and the results shown in **Fig. 3.4**.

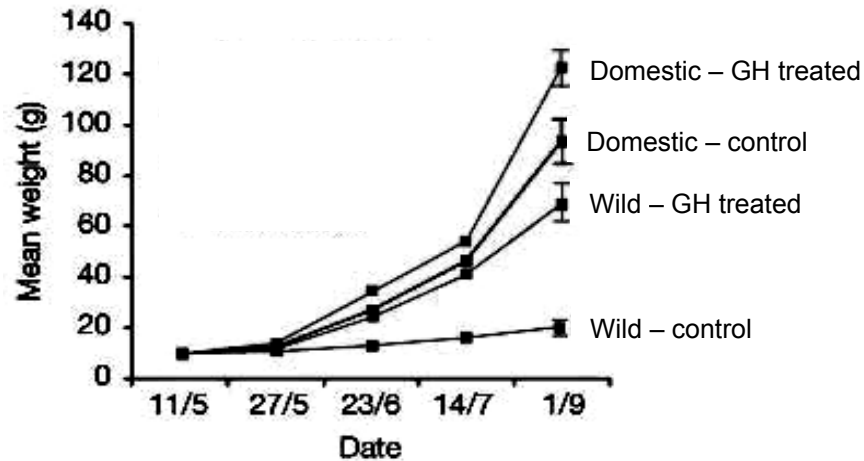


Fig. 3.3

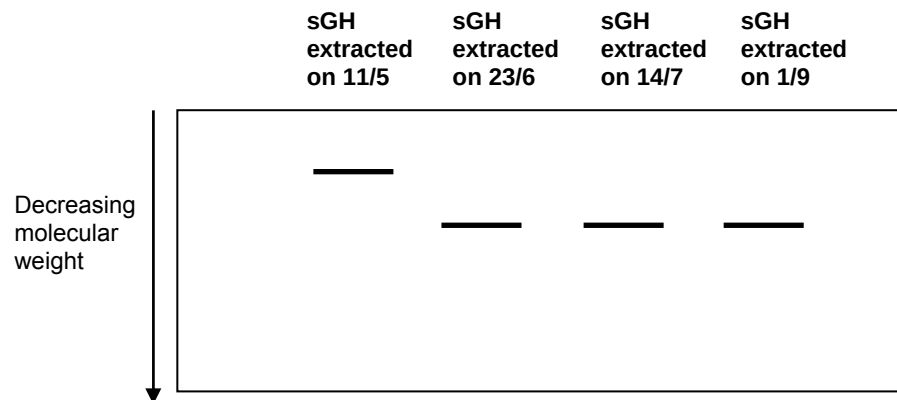


Fig. 3.4

(d) Using the information given in **Fig 3.3**,

(i) describe the effect of the growth hormone on salmon.

Domestic GH-treated reached maximum weight of 130g while Domestic-control reached maximum weight of 90g /

Domestic GH-treated had increase in weight gain of 120g while Domestic-control had increase in weight gain of 80g /

Wild GH-treated reached maximum weight of 65g while Wild-control reached maximum weight of 15g /

Wild GH-treated had increase in weight gain of 55g while Wild-control had increase in weight gain of 5g;

[2]

- (ii) suggest why there is a difference in weight gain between domestic-GH treated and wild-GH treated fish.

Other factors such as environment / nutrients, could play a role in weight gain;

[1]

- (e) With reference to both **Fig. 3.3** and **Fig. 3.4**, suggest the process sGH undergoes before exerting its effect on weight gain.

Pre-hormone synthesized on 11/5 undergoes post-translational / structural modification / had some peptide sequences cleaved off to form mature / functional hormone;

[2]

[Total: 15]

Section B

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4 **(a)** Describe how the micropropagation technique is used in cloning of plants. [6]
1. Preparing sterile media containing nutrients and hormones needed for plant growth / antibiotics / herbicides for selection of successfully transformed cells;
 2. Insertion of transgene via microprojectile bombardment / electroporation / *Agrobacterium tumefaciens*-mediated transformation;
 3. Explant containing meristematic cells is selected and surface sterilized before growing on medium;
 4. As callus increases in size, pieces of callus is sliced off and grown on new medium composition;
 5. By adjusting concentration of plant hormones in growth medium, cells in callus can be induced to differentiate into roots and shoots;
 6. Plantlets are removed from agar medium and transplanted to sterile soil for further growth;

- (b) Explain, with specific examples, how genetic engineering can improve the quality and yield of crop plants in solving the demand for food in the world. [9]

Candidate must elaborate on at least 3 specific e.g. which must cover both improvements in 'quality' and 'yield'. If candidate only elaborated in one aspect, maximum of 6 marks will be awarded.

Examples for improvement in yield of crop plants (any 2 e.g.)

Pest-resistant transgenic plants

1. Transgenic plants containing the Bt gene will produce the protoxin which is ingested by the insect; Protoxin is cleaved in the insect gut and the active Bt toxin is released;
2. Protein binds to cell membranes and causes them to be permeable causing gut cells of insect to lyse, eventually killing insects;

Herbicide-resistant transgenic plants

3. Glyphosate is a herbicide used to get rid of weeds which compete with crops for soil nutrients;
4. Inhibits 5-enolpyruvylshikimate-3-phosphate (EPSP) synthase needed for biosynthesis of essential amino acids;
5. Transgenic plants contain mutated gene which codes for mutant EPSP synthase that is not inhibited by glyphosate;

Extension of shelf life of produce

6. Transgenic plants contain antisense polygalacturonase mRNA;
7. Results in the formation of mRNA-antisense RNA hybrid, which prevents the polygalacturonase mRNA from being translated;
8. Enzyme will not be synthesized, thus ensures fruits are not damaged / too ripe / soft when it reaches consumer in market place;

Examples for improvement in quality of crop plants (must have this e.g.)

Improving nutritional value of crops

9. Golden rice is a transgenic plant that three beta-carotene biosynthesis genes;
10. Produces yellow rice grains containing beta-carotene, which is a precursor to vitamin A;
11. Prevent vitamin A deficiency in world's population that depends on rice as a staple food;

(c) Discuss the ethical and social implications of genetically modified organisms. [5]

Social implications (maximum 3 marks)

1. Transfer of antibiotic resistance markers to pathogenic microorganisms which may result in increase in resistance to clinically important antibiotics;
2. Probability of introducing novel allergens as GM foods may contain proteins introduced from sources people are allergic to;
3. Possibility of GM food being toxic or carcinogenic as they may cause over-expression of other proteins;
4. Monopolistic behaviour of biotechnology companies as terminator gene is likely to be inserted into many GMO seeds, causing second generation seeds to be sterile;
5. Scientists have raised concerns about innovations in research that are not shared, raising fears that world food production may be dominated by a few large biotechnology companies;
6. Increasing dependence of developing nations on industrialized nations;
7. Impact on international trade as Europe has been much more hesitant than the United States in accepting GM products in processed food;

Ethical issues (maximum 2 marks)

8. Tampering with nature as it is going against the natural way of life;
9. Lack of mandatory food labelling in some countries;
10. Religious groups are concerned that GM foods might contain genes from animals prohibited by their religion;